

Second Normal Form (2NF) Relational Database Design

> **Definition:** A relation is in **Second Normal Form (2NF)** if it is in **1NF** and **no non-prime attribute is partially dependent on any candidate key** of the relation. Every non-key attribute must be **fully functionally dependent** on the primary key.

1. Detailed Technical Explanation

Functional Dependency Concepts:

- **Full Functional Dependency:** $X \twoheadrightarrow Y$ is a full functional dependency if removal of any attribute A from X means the dependency no longer holds.

Conversion Example: 1NF to 2NF

Relation R(Student_ID, Course_ID, Student_Name, Course_Fee)

Decomposition into 2NF:

1. **STUDENT** (Student_ID, Student_Name)

2. Core Concepts & Memory Keywords

- **Non-prime Attribute:** An attribute that is not part of any candidate key.

3. Must-Write Points for Exams

- 2NF is only relevant when the relation has a **composite candidate key** (candidate key containing 2 or more attributes).

4. Quick Recall Flow
